



How the Human Heart Pumps Blood - Interactive Lesson

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Best for: Upper elementary to middle school learners

Estimated time: 15-25 minutes

Learning goal: Explain the path blood takes through the heart, lungs, and body, and describe how the heart's rhythm keeps blood moving.

Success criteria: By the end, you can trace the blood-flow sequence, define key terms, and measure your pulse safely.

The Question(TM)

How does the heart keep blood moving through my entire body every second of my life?

Before you learn: Safe pulse check

Try to find your pulse. Use two fingers - not your thumb - and press gently.

- Wrist option: Place two fingers on the thumb-side of your wrist.
- Neck option: Place two fingers gently on one side of your neck, beside your windpipe. Do not press hard, and never press both sides of the neck at the same time.
- Alternative: If you cannot easily feel a pulse, place a hand lightly over your chest, notice your breathing and heartbeat, or simply skip the pulse activity and use the lesson diagram.

Can you feel a beat? That pulse is evidence that your heart is pushing blood through your arteries.

Think first: Is the pulse the blood moving, the heart squeezing, or both? Keep your answer in mind as you read.

Simple Answer(TM)

Your heart is a strong muscular pump that squeezes and relaxes again and again - about 100,000 times a day - to move blood through two main loops:

- Pulmonary circulation: blood travels from the heart to the lungs and back.
- Systemic circulation: blood travels from the heart to the rest of the body and back.

Blood carries oxygen and nutrients to your cells and carries away wastes, such as carbon dioxide.

Big idea: The right side of the heart sends blood to the lungs. The left side sends blood to the body.

QRU Translation(TM)

Think of your heart as a two-story house with four rooms:

- The top rooms are the atria. They receive blood.
- The bottom rooms are the ventricles. They pump blood out.
- Valves are one-way doors that open and close because of pressure changes, helping blood move forward instead of backward.
- Arteries carry blood away from the heart.
- Veins carry blood back to the heart.

Important vocabulary you will use:

- Vena cava: the large veins that bring oxygen-poor blood from the body to the right atrium.
- Pulmonary arteries: vessels that carry oxygen-poor blood from the right ventricle to the lungs.
- Pulmonary veins: vessels that carry oxygen-rich blood from the lungs to the left atrium.
- Aorta: the large artery that carries oxygen-rich blood from the left ventricle to the body.

Artery/vein exception to remember:

Most arteries carry oxygen-rich blood, and most veins carry oxygen-poor blood. But the pulmonary arteries and pulmonary veins are exceptions. Arteries are named because they carry blood away from the heart, not because of how much oxygen they carry.

The heart also has a built-in electrical system. A small group of cells often called the heart's natural pacemaker starts each heartbeat. The signal spreads through the atria first, then to the ventricles, so the chambers squeeze in a coordinated rhythm instead of all at once.

Two key heartbeat words:

- Systole: the heart muscle squeezes and pushes blood forward.
- Diastole: the heart muscle relaxes and fills with blood.

Quick visual map:

text

KEY:

[O2-poor] = oxygen-poor blood going to the lungs

[O2-rich] = oxygen-rich blood going to the body

PULMONARY CIRCULATION: heart ? lungs ? heart

BODY

? [O2-poor] through vena cava

RIGHT ATRIUM

? through tricuspid valve

RIGHT VENTRICLE

? through pulmonary valve

PULMONARY ARTERIES

? [O2-poor] to lungs

LUNGS

? oxygen enters blood; carbon dioxide leaves blood

PULMONARY VEINS

? [O2-rich] back to heart

LEFT ATRIUM

SYSTEMIC CIRCULATION: heart ? body ? heart

LEFT ATRIUM

? through mitral valve

LEFT VENTRICLE

? through aortic valve

AORTA

? [O₂-rich] through arteries

BODY

? cells use oxygen; blood picks up carbon dioxide

VENA CAVA

? [O₂-poor] back to heart

RIGHT ATRIUM

Diagram alternative text: Oxygen-poor blood returns from the body through the vena cava to the right side of the heart, travels through the pulmonary arteries to the lungs, returns oxygen-rich through the pulmonary veins to the left side of the heart, and is pumped through the aorta to the body.

Mini-check: Point to the map and answer:

1. Which side of the heart sends blood to the lungs?
2. Which side sends blood to the body?
3. Why are pulmonary arteries called arteries even though they carry oxygen-poor blood?

What's Happening Inside?(TM)

Blood follows a one-way route through the heart:

1. Body ? Right Atrium

Oxygen-poor blood returns from the body through the superior and inferior vena cava and enters the right atrium.

2. Right Atrium ? Right Ventricle

Blood flows through the tricuspid valve into the right ventricle. This valve helps stop blood from flowing backward.

3. Right Ventricle ? Lungs

The right ventricle squeezes and pumps blood through the pulmonary valve into the pulmonary arteries. These arteries carry oxygen-poor blood to the lungs. This is part of pulmonary circulation.

4. Lungs ? Left Atrium

In the lungs, blood releases carbon dioxide and picks up oxygen. Oxygen-rich blood returns to the heart through the pulmonary veins and enters the left atrium.

5. Left Atrium ? Left Ventricle

Blood flows through the mitral valve into the left ventricle.

6. Left Ventricle ? Body

The left ventricle squeezes and pumps blood through the aortic valve into the aorta. From there, oxygen-rich blood travels through arteries to the body. This is part of systemic circulation.

7. Body ? Heart Again

Body cells use oxygen and nutrients and add wastes such as carbon dioxide to the blood. Oxygen-poor blood returns through veins to the vena cava, and the cycle repeats.

Sequence challenge: Put these in the correct order without looking. Then check with the map.

text

Aorta

Right atrium

Lungs

Left ventricle

Vena cava

Pulmonary arteries

Left atrium

Right ventricle

Pulmonary veins

Body

Correct sequence:

text

Body ? Vena cava ? Right atrium ? Right ventricle ? Pulmonary arteries ? Lungs ? Pulmonary veins ?
Left atrium ? Left ventricle ? Aorta ? Body

What makes the blood move?

The heart does not push blood by "stirring" it. It moves blood by coordinated squeezing:

- During diastole, chambers relax and fill.
- During systole, chambers squeeze and push blood forward.
- Valves open when pressure pushes blood in the correct direction.
- Valves close when blood tries to move backward.

That "lub-dub" heartbeat sound is connected to valves closing as blood moves through the heart.

Pulse investigation

1. Sit calmly for one minute.
2. Find your wrist pulse safely.
3. Count beats for 15 seconds.
4. Multiply by 4 to estimate beats per minute.
5. If it is safe and appropriate for you, do 20 gentle jumping jacks or march in place for 30 seconds.
6. Count again.

Record your results:

Condition	Beats in 15 seconds	Estimated beats per minute
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Resting		
After gentle movement		

Reflect: What changed? Why might muscles need more blood during movement?

Check for understanding

1. Trace it: Complete the missing steps.

Body ? vena cava ? ? right ventricle ? pulmonary arteries ? ? pulmonary veins ? left atrium ? ? aorta ? body

2. Define it:

- Artery:
- Vein:
- Valve:
- Pulse:

3. Explain the exception:

Why can the pulmonary artery carry oxygen-poor blood and still be called an artery?

4. Connect rhythm to pumping:

What would happen if the atria and ventricles squeezed in a disorganized way instead of a coordinated rhythm?

Answer key

1. Right atrium; lungs; left ventricle.

2. Artery: carries blood away from the heart. Vein: carries blood back to the heart. Valve: a one-way flap that helps blood move forward. Pulse: the beat you feel in an artery when the heart pumps blood.

3. It is called an artery because it carries blood away from the heart; pulmonary arteries are an oxygen-level exception.

4. Blood would not move as efficiently because the chambers need to fill and squeeze in the correct order.

Exit ticket

Before you finish, say or write one complete sentence for each:

- The right side of the heart pumps blood to .
- The left side of the heart pumps blood to .
- One way to measure pulse safely is .
- One new heart vocabulary word I can explain is .

Continue learning at QRU:

